Explaining the Link Between Physical Activity & Breathing

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 3 October 2024

Explaining the Link Between Physical Activity & Breathing: Extended

Extended Tier Only

- Frequency and depth of breathing **increase when exercising**
- This is because muscles are working harder and aerobically respiring more and they **need more oxygen to be delivered to them** (and carbon dioxide removed) to keep up with the energy demand
- If they cannot meet the energy demand they will also **respire anaerobically**, producing **lactic acid**
- After exercise has finished, the lactic acid that has built up in muscles needs to be **removed** as it **lowers the pH of cells** and can **denature enzymes** catalysing cell reactions
- It can only be removed by combining it with oxygen - this is known as **'repaying the oxygen debt'**
- This can be tested by seeing how long it takes after exercise for the breathing rate and depth to return to normal - **the longer it takes, the more lactic acid produced during exercise and the greater the oxygen debt that needs to be repaid**

Mechanism for increasing breathing during exercise

- The **rate of respiration** increases in muscle cells when exercising heavily
- CO_2 is a **product** of aerobic respiration, so **CO_2 levels increase** in the muscle cells
- This CO_2 diffuses out of the cells into the **blood plasma**
  - CO_2 in solution causes a slight drop in pH so the blood becomes slightly more acidic
- The blood flows around the circulatory system and passes to the brain where the increased carbon dioxide levels are detected by **chemoreceptors** in the brain
  - Chemoreceptors are cells that detect chemical changes in the body
  - They can detect changes in blood gas levels, as well as changes in pH

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

intercostal muscles to **increase the rate and depth of muscle contraction**
- The **rate of inspiration increases**, along with the the **volume of air** moved in and out with each breath
- The result is **greater absorption of oxygen** and **removal rate of carbon dioxide**
  - This supports the **increased rate of respiration** in the exercising muscle cells

Examiner Tips and Tricks

Be sure not to confuse the terms **respiration** and **breathing** in this topic:

- Respiration is a series of chemical reactions that releases energy from food
- Breathing is the mechanism for moving air in and out of the lungs; this supports the chemical reactions of respiration

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.